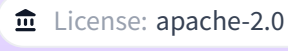


 **cl-nagoya/ruri-base** 

 like 10

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 Sentence Similarity  Safetensors  cl-nagoya/ruri-dataset-ft  Japanese

 bert  feature-extraction  arxiv:2409.07737  License: apache-2.0



 **Model card**  Files  xet  Community

Downloads last month

266,933



 **Safetensors** 

Model size 111M params

Tensor type BF16

 Files info

 **Inference Providers** NEW

 Sentence Similarity

This model isn't deployed by any Inference Provider.

 Ask for provider support

 **Model tree for cl-nagoya/ruri-base**

Base model [tohoku-nlp/bert-base-japanese-v3](#)

 Finetuned [cl-nagoya/ruri-pt-base](#)

 Finetuned (4) [this model](#)

 Finetunes [1 model](#)

 **Dataset used to train cl-nagoya/ruri-base**

 **Collection including cl-nagoya/ruri-base**

Ruri: Japanese General Text Embeddings  Collection

26 items • Updated Apr 18 •  14

 **Ruri: Japanese General Text Embeddings**

Notes: v3 models are out!

We recommend using the following v3 models going forward.

ID	#Param.	Max Len.	Avg. JMTEB
cl-nagoya/ruri-v3-30m	37M	8192	74.51
cl-nagoya/ruri-v3-70m	70M	8192	75.48
cl-nagoya/ruri-v3-130m	132M	8192	76.55
cl-nagoya/ruri-v3-310m	315M	8192	77.24

🔗 Usage

🔗 Direct Usage (Sentence Transformers)

First install the Sentence Transformers library:

```
pip install -U sentence-transformers fugashi sentencepiece unidic-lite
```

Then you can load this model and run inference.

```
import torch.nn.functional as F
from sentence_transformers import SentenceTransformer

# Download from the 🤗 Hub
model = SentenceTransformer("cl-nagoya/ruri-base")

# Don't forget to add the prefix "クエリ: " for query-side or "文章: " for passage-side
sentences = [
    "クエリ: 瑠璃色はどんな色?",
    "文章: 瑠璃色（るりいろ）は、紫みを帯びた濃い青。名は、半貴石の瑠璃（ラピスラズリ、英: lapis lazuli）",
    "クエリ: ワシやタカのように、鋭いくちばしと爪を持った大型の鳥類を総称して「何類」というでしょう?",
    "文章: ワシ、タカ、ハゲワシ、ハヤブサ、コンドル、フクロウが代表的である。これらの猛禽類はリンネによって分類された。"
]
```

```

]

embeddings = model.encode(sentences, convert_to_tensor=True)
print(embeddings.size())
# [4, 768]

similarities = F.cosine_similarity(embeddings.unsqueeze(0), embeddings.unsqueeze(1))
print(similarities)
# [[1.0000, 0.9421, 0.6844, 0.7167],
#  [0.9421, 1.0000, 0.6626, 0.6863],
#  [0.6844, 0.6626, 1.0000, 0.8785],
#  [0.7167, 0.6863, 0.8785, 1.0000]]

```

[Benchmarks](#)

[JMTEB](#)

Evaluated with [JMTEB](#).

Model	#Param.	Avg.	Retrieval	STS	Classification	Reranking	Clustering	PairC
cl-nagoya/sup-simcse-ja-base	111M	68.56	49.64	82.05	73.47	91.83	51.79	
cl-nagoya/sup-simcse-ja-large	337M	66.51	37.62	83.18	73.73	91.48	50.56	
cl-nagoya/unsup-simcse-ja-base	111M	65.07	40.23	78.72	73.07	91.16	44.77	
cl-nagoya/unsup-simcse-ja-large	337M	66.27	40.53	80.56	74.66	90.95	48.41	
pkshatech/GLuCoSE-base-ja	133M	70.44	59.02	78.71	76.82	91.90	49.78	
sentence-transformers/LaBSE	472M	64.70	40.12	76.56	72.66	91.63	44.88	

Model	#Param.	Avg.	Retrieval	STS	Classification	Reranking	Clustering	PairC
intfloat/multilingual-e5-small	118M	69.52	67.27	80.07	67.62	93.03	46.91	
intfloat/multilingual-e5-base	278M	70.12	68.21	79.84	69.30	92.85	48.26	
intfloat/multilingual-e5-large	560M	71.65	70.98	79.70	72.89	92.96	51.24	
OpenAI/text-embedding-ada-002	-	69.48	64.38	79.02	69.75	93.04	48.30	
OpenAI/text-embedding-3-small	-	70.86	66.39	79.46	73.06	92.92	51.06	
OpenAI/text-embedding-3-large	-	73.97	74.48	82.52	77.58	93.58	53.32	
Ruri-Small	68M	71.53	69.41	82.79	76.22	93.00	51.19	
Ruri-Base (this model)	111M	71.91	69.82	82.87	75.58	92.91	54.16	
Ruri-Large	337M	73.31	73.02	83.13	77.43	92.99	51.82	

[🔗](#) **Model Details**

[🔗](#) **Model Description**

- **Model Type:** Sentence Transformer
- **Base model:** [cl-nagoya/ruri-pt-base](#)
- **Maximum Sequence Length:** 512 tokens
- **Output Dimensionality:** 768
- **Similarity Function:** Cosine Similarity

- **Language:** Japanese
- **License:** Apache 2.0
- **Paper:** <https://arxiv.org/abs/2409.07737>

🔗 Full Model Architecture

```
SentenceTransformer(  
  (0): Transformer({'max_seq_length': 512, 'do_lower_case': False}) with Transform  
  (1): Pooling({'word_embedding_dimension': 768, 'pooling_mode_cls_token': False,  
)
```

🔗 Framework Versions

- Python: 3.10.13
- Sentence Transformers: 3.0.0
- Transformers: 4.41.2
- PyTorch: 2.3.1+cu118
- Accelerate: 0.30.1
- Datasets: 2.19.1
- Tokenizers: 0.19.1

🔗 Citation

```
@misc{  
  Ruri,  
  title={{Ruri: Japanese General Text Embeddings}},  
  author={Hayato Tsukagoshi and Ryohei Sasano},  
  year={2024},  
  eprint={2409.07737},  
  archivePrefix={arXiv},  
  primaryClass={cs.CL},  
  url={https://arxiv.org/abs/2409.07737},
```

```
}
```

[🔗](#) License

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